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Clinical application of sequencing-based methods for parallel preimplantation genetic testing for mitochondrial DNA disease and aneuploidy

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Objective: To validate and apply a strategy permitting parallel preimplantation genetic testing (PGT) for mitochondrial DNA (mtDNA) disease and aneuploidy (PGT-A).
Design: Preclinical test validation and case reports.
Setting: Fertility centers. Diagnostics laboratory.
Patients: Four patients at risk of transmitting mtDNA disease caused by m.8993T>G (Patients A and B), m.10191T>G (Patient C), and m.3243A>G (Patient D). Patients A, B, and C had affected children. Patients A and D displayed somatic heteroplasmy for mtDNA mutations.
Interventions: Embryo biopsy, genetic testing, and uterine transfer of embryos predicted to be euploid and mutation-free.
Main Outcome Measures: Test accuracy, treatment outcomes, and mutation segregation.
Results: Accuracy of mtDNA mutation quantification was confirmed. The test was compatible with PGT-A, and half of the embryos tested were shown to be aneuploid (16/33). Mutations were detected in approximately 40% of embryo biopsies from Patients A and D (10/24) but in none from Patients B and C (n = 29). Patients B and C had healthy children following PGT and natural conception, respectively. The m.8993T>G mutation displayed skewed segregation, whereas m.3243A>G mutation levels were relatively low and potentially impacted embryo development.
Conclusions: Considering the high aneuploidy rate, strategies providing a combination of PGT for mtDNA disease and aneuploidy may be advantageous compared with approaches that consider only mtDNA. Heteroplasmic women had a higher incidence of affected embryos than those with undetectable somatic mutant mtDNA but were still able to produce mutation-free embryos. While not conclusive, the results are consistent with the existence of mutation-specific segregation mechanisms occurring during oogenesis and possibly embryogenesis. (Fertil Steril® 2021;115:1521–32. ©2021 by American Society for Reproductive Medicine.)
El resumen está disponible en Español al final del artículo.

Key Words: Preimplantation genetic testing, mitochondrial DNA disease, next-generation sequencing, embryos, mitochondria

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Mitochondrial dysfunction caused by pathogenic mutations in nuclear DNA (nDNA) or mitochondrial DNA (mtDNA) can result in severe multisystemic diseases (1). These cannot be cured, and treatment is limited to symptom management (2). Various reproductive options for reducing disease recurrence are available to women at risk of transmitting mitochondrial disease. These include in vitro fertilization (IVF) with donor oocytes, preimplantation genetic testing (PGT), and prenatal diagnosis (PND) (3, 4). PGT and PND offer the advantage that couples can conceive healthy and genetically related children. However, if an affected fetus is identified by

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PND, couples face the difficult decision of continuing or terminating the pregnancy. A significant advantage of PGT is the low risk of carrying a fetus with mitochondrial disease because only embryos predicted to be free of a clinical phenotype are considered for uterine transfer (5).

PGT for mitochondrial disease caused by nDNA mutations involves assessing the presence or absence of the causative mutation. PGT for disease resulting from mtDNA mutations also requires a quantitative component to determine frequencies of wild-type and mutant mtDNA molecules. This is crucial because most mtDNA mutation carriers are heteroplasmic (i.e., cells contain wild-type and mutant mitochondria). Disease only occurs when mutant mitochondria proportions exceed critical mutation-specific thresholds (6). Consequently, not only mutation-free embryos but also those with low mutation levels, unlikely to cause disease, may be eligible for transfer (7). Mutation levels can differ markedly between sibling embryos, presumably a consequence of the mitochondrial bottleneck occurring during female gametogenesis (6, 8). Consequently, women affected by mtDNA disease and asymptomatic women carrying pathogenic mtDNA mutations can benefit from PGT, as both are at risk of conceiving clinically affected children (9).

Methods employed for PGT for mtDNA disease are commonly based on restriction fragment length polymorphism analysis or on amplification refractory mutation system techniques, whereby a mutation is targeted by an allele-specific primer or probe or by a restriction endonuclease, respectively, and then quantified using techniques such as electrophoresis or quantitative polymerase chain reaction (PCR) (10–15). These methods are technically challenging and difficult to combine with other diagnostic or screening procedures. For instance, combination with testing for aneuploidy or gender is problematic. Aneuploidy, however, is extremely common in human embryos, causing implantation failure, miscarriage, and aneuploid pregnancies (16–20). Furthermore, the transfer of a male embryo, if available and legally permitted, may be considered to avoid disease transmission to future generations (7). This is particularly relevant if entirely mutation-free embryos cannot be identified and embryos with subclinical mutation levels are transferred.

We designed a novel method of PGT for mtDNA disease, which can easily be combined with PGT for aneuploidy (PGT-A). Because most PGT-A methods require whole-genome amplification (WGA) to enrich the genetic starting material, we established a next-generation sequencing (NGS) method, which uses WGA products for the detection and quantification of mtDNA mutations in embryo specimens. We present extensive validation of the PGT approach and its clinical application in embryo specimens from 4 patients, referred either for Leigh syndrome caused by m.8993T>G or m.10191T>G mutations or for mitochondrial myopathy caused by the m.3243A>G mutation. We further provide information concerning the segregation of different mtDNA mutations during oogenesis and embryogenesis.

MATERIALS AND METHODS

Patient Details

Pedigrees of Patients A, B, C, and D are shown in Figure 1. None of the patients' families had a known history of mtDNA disease. For Patients A, B, and C, the possibility of disease risk became apparent after children with Leigh syndrome were born. Patient A's daughter and Patient B's son were effectively homoplasmic (approximately 100%; blood) and heteroplasmic (approximately 96%; blood) for the m.8993T>G mutation, respectively. Patient C's daughter was heteroplasmic (approximately 78%; muscle) for the m.10191T>C mutation. Patients A, B, and C and Patient B's mother underwent genetic testing. A mutation level of approximately 56% was identified in Patient A's blood. Conversely, blood, urine, and cheek swabs of Patient B and her mother and the blood and urine of Patient C were free of the mutations observed in their children. Patient D had no child and was undergoing PGT due to the presence of approximately 28% heteroplasmy of the m.3243A>G mutation (muscle), which caused her to suffer from mitochondrial myopathy, resulting in ptosis and restricted eye movement. Patient D's mother and brother were not tested. However, both presented with hearing and vision loss, symptoms frequently associated with the m.3243A>G mutation (21, 22).

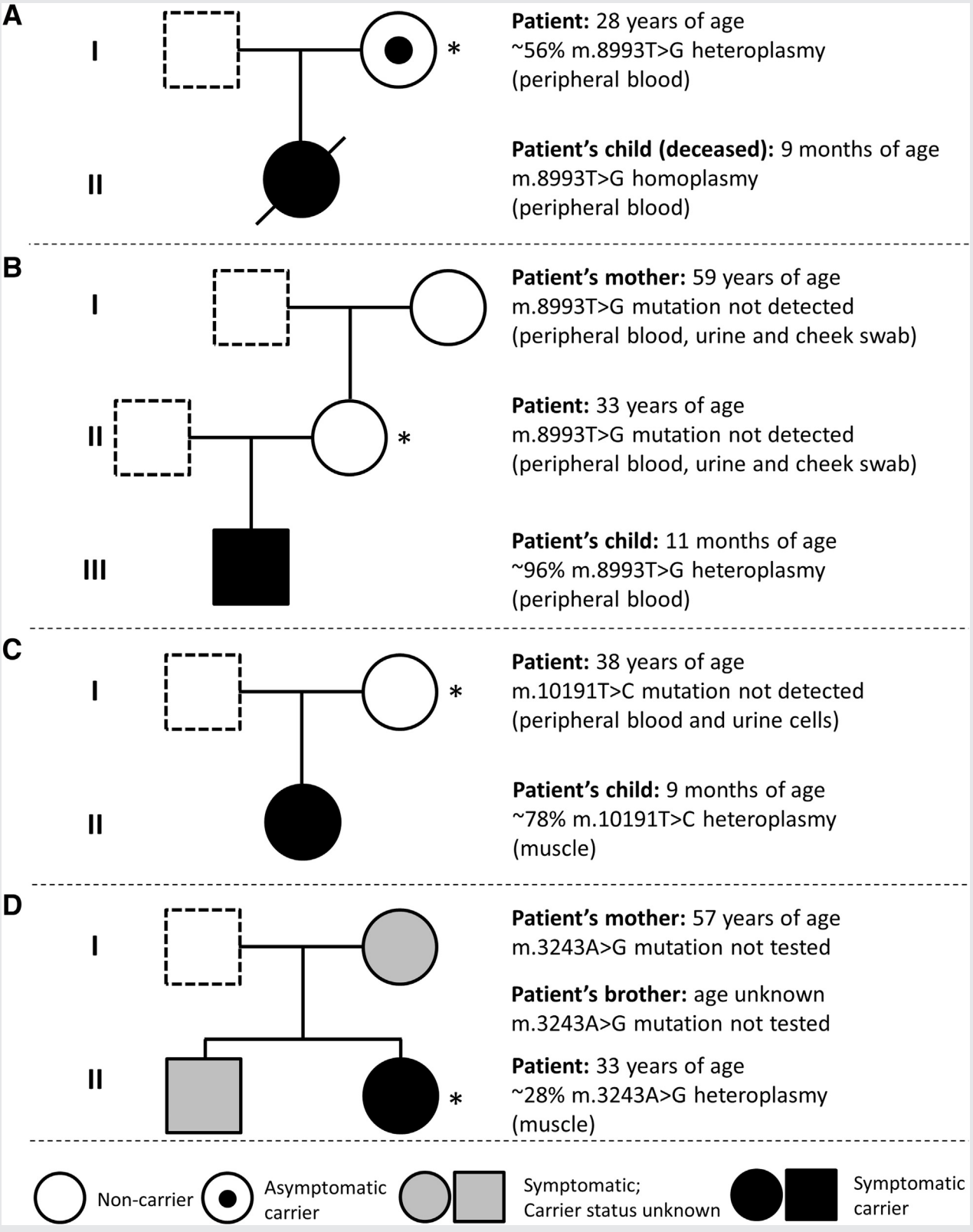
Patients were provided with treatment descriptions and information on method limitations, risks, and potential benefits and were counseled and consented before treatment initiation. Patient consent also was obtained for the use of their own or their children's genomic DNA (gDNA) and cell samples for method development and validation. Furthermore, patient consent was obtained for the anonymous publication of the results of their treatment cycles. Because all patient information was fully de-identified, and data were produced during the course of routine treatment, the requirement for a specific Institutional Review Board approval was waived.

Techniques Used

The SurePlex DNA Amplification System (Illumina, San Diego, CA, USA) was employed for WGA and used as recommended by the supplier. PGT-A was performed by array comparative genomic hybridization (aCGH) or NGS according to Illumina's 24sure or VeriSeq protocols, respectively. For aCGH, using 24sure microarrays, hybridization areas were identified and captured using an InnoScan 700 laser scanner (Innopsys, Chicago, IL, USA). For NGS, using the VeriSeq PGS Kit, the MiSeq Sequencing System (Illumina) was used. All aCGH and sequencing outputs were analyzed using Bluefuse Multi Software (Illumina).

For PGT for mtDNA disease, patient-specific mutation loci were enriched by conventional PCR using flanking primers (m.8993: 5′-CCCTAGCCCACTTCTTACCA-3′ and 5′-GGTGGCCTGCAGTAATGTTAG-3′; m.10191: 5′-ATCCACCCCTTACGAGTGC-3′ and 5′-GGGTAAAAGGAGGGCAATTT-3′; m.3243: 5′-ACCCAAGAACAGGGTTTGT-3′ and 5′-TGGCCATGGGTATGTTGTTA-3′). Each PCR sample

FIGURE 1



Pedigrees of patients referred for PGT. Pedigrees **A**, **B**, **C**, and **D** refer to Patients **A**, **B**, **C**, and **D**, respectively. Roman numerals indicate family generations. Stars mark patients. DNA sources are given in brackets. The specified ages refer to patients', relatives', and children's ages at the time when genetic testing was performed. PGT = preimplantation genetic testing.

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consisted of 0.5 μL WGA product, 12.49 μL nuclease-free water (Roche Diagnostics, Indianapolis, IN, USA), 1.5 μL HotMaster Taq DNA Buffer with Magnesium (Five Prime Therapeutics, San Francisco, CA, USA), 0.12 μL primer pool (100 μM; Eurogentec, Liège, Belgium), 0.3 μL dNTP Mix (10 mM; ThermoFisher Scientific, Waltham, MA, USA), and 0.09 μL HotMaster Taq DNA Polymerase (Five Prime). PCRs were performed at following conditions: 96.0°C for 1 minute, followed by 35 repeats of 94.0°C for 15 seconds, 61.0°C (m.8993) or 57.0°C (m.10191 and m.3243) for 15 seconds and 65.0°C for 45 seconds, followed by 65.0°C for 2 minutes. Reagents, sequencing equipment, and data analysis tools for mtDNA variant detection and quantification were obtained from ThermoFisher Scientific (Waltham, MA, USA), and all procedures were performed according to the supplier's recommendations. PCR products were processed using the Ion Plus Fragment Library and Ion PGM Template OT2 200 Kits. The resulting libraries were sequenced on the Ion Personal Genome Machine System using the Ion PGM Sequencing 200 Kit v2. The Torrent Browser and inbuilt Torrent Suit Software were used for data processing. The Torrent Variant Caller plugin was used to detect and quantify mtDNA mutations.

Validation of the Method

The PGT for mtDNA disease method was validated using gDNA and cell samples from cheek swabs or blood collected from patients, their affected children, and control individuals. The DNA Isolation kit DDK-50 (Cell Projects, Ltd., Harrietsham, United Kingdom) and the BuccalAmp DNA Extraction Kit (Epicentre, Madison, WI, USA) were used to obtain gDNA and buccal cells from cheek swabs, respectively. The QIAamp DNA Mini Kit (Qiagen, Hilden, Germany) was used to extract gDNA from the blood. Ficoll-Paque Plus (GE Healthcare Life Sciences, Chicago, IL, USA) was used to separate lymphocytes. Buccal cells and lymphocytes were isolated manually by micromanipulation and washed in Dulbecco's phosphate-buffered saline (ThermoFisher)/0.1% polyvinyl alcohol (Sigma-Aldrich, St. Louis, MO, USA).

The method was initially tested on gDNA from 2 individuals, homoplasmic for either the m.8993 wild-type (T) or mutant (G) variant. To test method accuracy and sensitivity and to ensure that WGA or amplification of the mutation site did not distort variant ratios, sample mixtures with 5.0%, 10.0%, 20.0%, 40.0%, 60.0%, and 80.0% mutant load were created by mixing homoplasmic gDNA specimens in appropriate amounts. Mutation site amplification was performed directly on sample mixtures and also on WGA products created from sample mixtures. Products were sequenced, and the obtained mutant loads were compared to expected loads.

To test protocol accuracy in samples consisting of single or few cells, equivalent to embryo biopsies, minute quantities of gDNA (approximately 100 pg) and cell samples (consisting of 1, 3, or 5 cells) from patients and affected children were analyzed. Again, samples were sequenced either after direct amplification of respective mutation loci or after WGA followed by conventional PCR. Of note, gDNA samples were isolated from the same cheek swabs or blood samples, as were individual buccal cells and lymphocytes. Results from gDNA and respective cell samples were compared with each other and with mutation levels previously established by reference laboratories.

Of note, for the direct amplification of mutation loci in cell samples, initial lysis was performed, whereby each sample was combined with 0.75 μL nuclease-free water, 1.25 μL DL-dithiothreitol solution (0.1 M; Sigma), and 0.5 μL sodium hydroxide solution (1.0 M; Sigma). Following incubation at 65°C for 10 minutes, 7.99 μL nuclease-free water and 2.5 μL Tricine (0.4 M; Roche Diagnostics) were added, followed by PCR components as described in **Techniques Used**.

Clinical Application of Established Method

IVF procedures, such as oocyte retrieval, embryo culture, biopsy, and transfer, were performed according to the clinics' standard protocols. Oocyte fertilization was performed by intracytoplasmic sperm injection. Biopsies were performed at the blastocyst stage (Patients A, B, and D) or the cleavage stage (Patient C). Biopsied embryos were vitrified, and those predicted to be unaffected were transferred in later cycles.

To maximize the information for patients concerning their disease transmission risk, testing was applied to unfertilized and abnormally fertilized (3 pronuclei) oocytes and to arrested embryos, which were either disaggregated into individual cells (providing information on mutation level uniformity within an embryo), split into halves, or tubed whole.

Patient B delivered a child following IVF with PGT. To confirm the PGT for mtDNA disease result, mutation analysis was performed in gDNA from cord blood and cheek swabs, sampled at 2 and 3 weeks of age.

Statistical Data Analysis

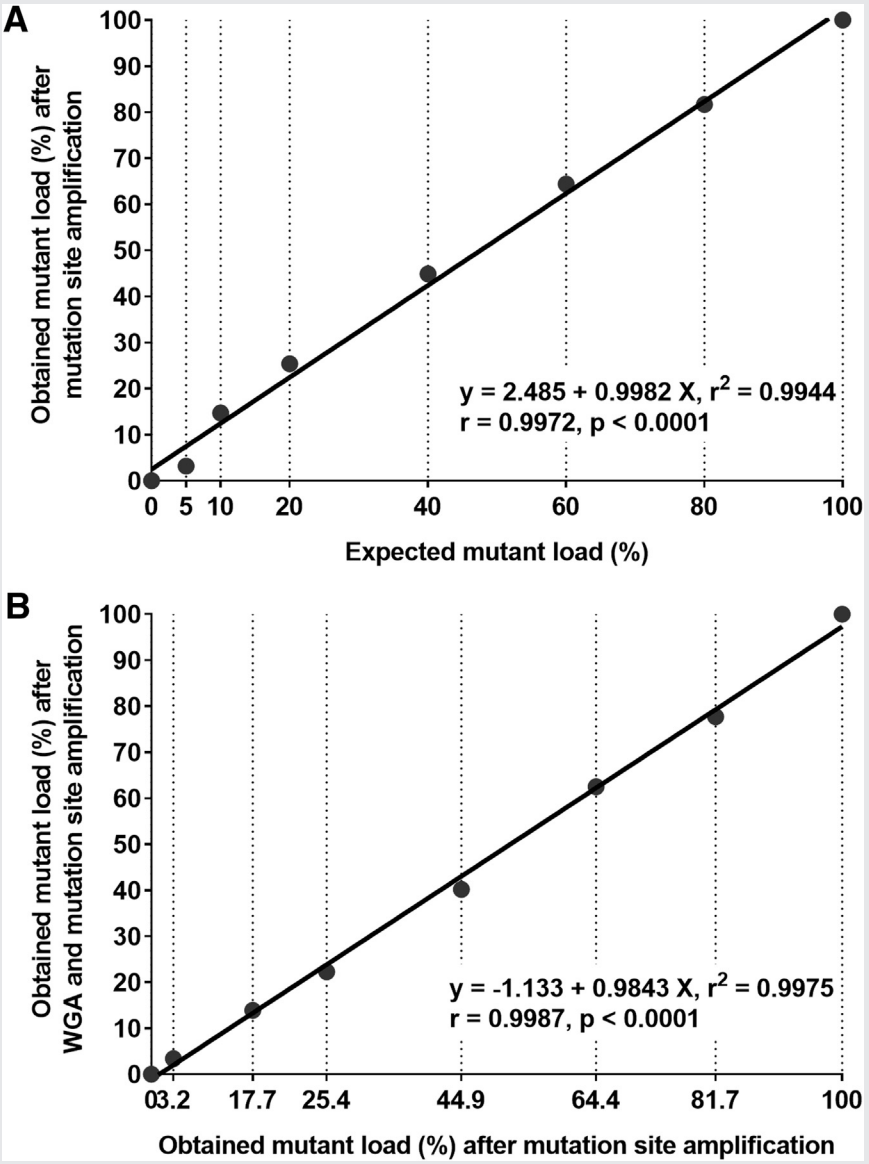
Data analysis was conducted using GraphPad Prism (GraphPad Software, San Diego, CA, USA). Linear regression analysis was performed to assess the linearity of obtained mutant loads. The Pearson's rank correlation coefficient was calculated to assess consistency between expected and obtained mutant loads. One-way ANOVA was used to compare mutation levels at different stages of preimplantation development. Tests were performed 2-tailed with a 95% confidence interval. A P value $<.05$ was considered significant.

RESULTS

Results of Method Validation

Figure 2 illustrates m.8993T>G mutant loads obtained from gDNA sample mixtures and pure (homoplasmic) samples after mutation site amplification alone (Fig. 2A) and after WGA combined with mutation site amplification (Fig. 2B). Numerical mutant loads are provided in Supplemental Table 1 (available online). Mutant loads obtained after direct mutation site amplification were nearly identical to expected ratios (mean difference ± SD, 2.4% ± 2.8%; r = 0.9972; $P<.0001$). A similar correlation (r = 0.9987; $P<.0001$) was observed when WGA was performed before mutation site amplification (mean difference ± SD, 1.8% ± 1.9%). Both experiments

FIGURE 2



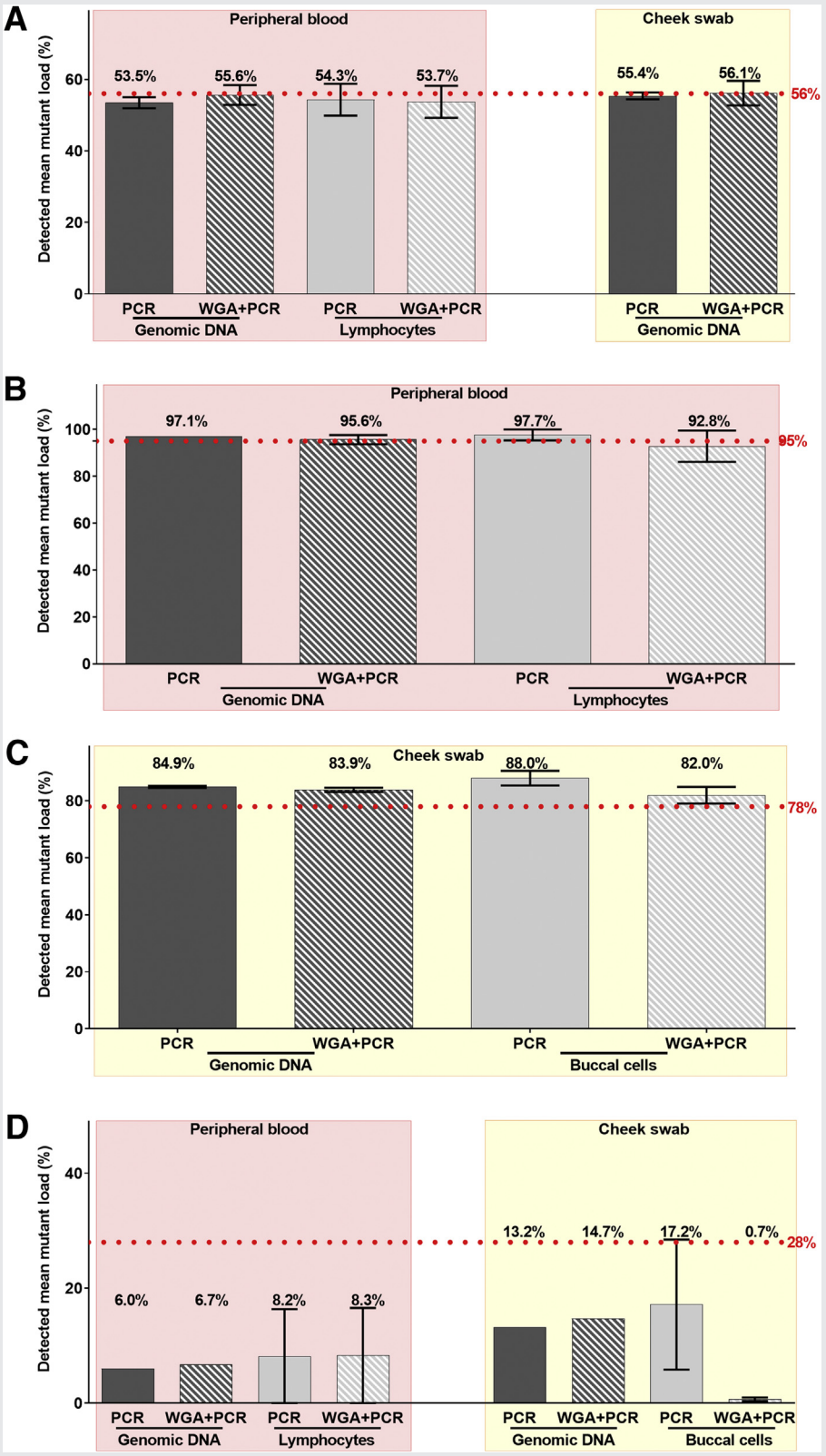
Correlation of obtained and expected mutant loads at position m.8993 (T>G) in gDNA sample mixtures. *Note*: r^2 indicates r-squared value after linear regression analysis; *y* indicates the equation of the best-fit line; r indicates Pearson's rank correlation coefficient with the corresponding *P* value. (**A**) Mutant loads obtained after direct mutation site amplification. (**B**) Mutant loads obtained after WGA followed by mutation site amplification.

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resulted in nearly perfect linear regressions ($r^2 = 0.9944$ and 0.9975). Of note, a homoplasmic result was obtained in all samples containing 0.0% or 100% mutant load. Furthermore, low-level heteroplasmy was confirmed in all samples containing the lowest expected mutant load of approximately 5.0%.

Figure 3 illustrates mean mutant loads detected in gDNA and cell samples from Patients A and D and children of Patients B and C. Mean mutant loads are shown after direct mutation site amplification and after WGA followed by amplification of respective mutation loci. Results from homoplasmic individuals (Patient A's child, mutant mtDNA; Patients B and C, wild-type mtDNA) are not illustrated as homoplasmy was confirmed in all samples. Numerical mutant loads are provided in Supplemental Table 2 (available online). Differences in mean mutant loads between samples subjected to direct mutation site amplification and those subjected to WGA before mutation site amplification were minimal (gDNA samples ± SD, 0.8% ± 0.8%; cell samples ± SD, 3.5% ± 5.8%). Mean differences were slightly higher in cell

FIGURE 3



Mean mutant loads detected in samples of heteroplasmic mtDNA mutation carriers. Mutant loads detected after direct mutation site amplification (PCR) and after WGA followed by mutation site amplification (WGA+PCR) in blood and/or cheek swab samples from (**A**) Patient A (m.8993T>G), (**B**) the son of Patient B (m.8993T>G), (**C**) the daughter of Patient C (m.10191T>C), and (**D**) Patient D (m.3243A>G). Error bars depict the standard error of the mean. Red dotted lines refer to mutation levels previously measured by reference laboratories. PCR = polymerase chain reaction; WGA = whole-genome amplification.

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samples, which likely reflects intercellular variation, ranging 0.0%–100%, 39.5%–100%, 75.2%–94.3%, and 0.0%–63.8% in Patient A, child of Patient B, child of Patient C, and Patient D, respectively.

Excellent agreement also was observed between gDNA samples and corresponding cell samples, differing on average by 1.5% ± 1.6% and 3.2% ± 4.9% after direct mutation site amplification and after WGA followed by PCR, respectively. The slightly higher difference in samples subjected to WGA was caused by a large variation between cheek swab gDNA and cell samples of Patient D (Fig. 3D).

Mean mutant loads in samples from individuals carrying the m.8993T>G mutation confirmed those previously reported by reference laboratories (Patient A, 54.3% ± 1.0% [blood] and 55.7% ± 0.5 [cheek swab] vs. approximately 56% [blood]; Patient A's daughter, 100% [blood] vs. 100% [blood]; Patient B's son. 95.2% ± 3.4% [blood] vs. approximately 95% [blood]). Mean mutant loads in samples from Patient C's daughter, who carries the m.10191T>C mutation, were slightly higher than the previously established mutation level (84.6 ± 2.5% [cheek swab] vs. approximately 78% [muscle]). Mean mutant loads measured in samples from Patient D, who carries the m.3243A>G mutation, were lower than previously established (7.3% ± 1.1% [blood] and 11.4% ± 7.3% [cheek swab] vs. approximately 28% [muscle]).

Results of Clinical Application

Results from all PGT procedures, including predicted embryo cytogenetic status and measured mtDNA mutation levels in oocyte/embryo specimens, are summarized in Supplemental Table 3 (available online).

Patient A underwent 2 consecutive cycles; 2 oocytes were collected in each cycle. Three oocytes were fertilized. One zygote developed to the blastocyst stage and was biopsied. The unfertilized oocyte and arrested embryos were tubed whole. Analysis of the m.8993T>G mutation was performed in all samples. PGT-A was performed in the biopsy and arrested embryos. The mutation was not detected in the biopsy and 1 embryo but was present at high levels in the unfertilized oocyte (91.2%) and the other embryo (88.7%). PGT-A revealed that the biopsy was aneuploid, with a karyotype of 48,XXY,+16. Consequently, the corresponding blastocyst was not transferred. PGT-A further revealed that the mutation-free, arrested embryo was euploid, whereas the arrested embryo carrying the mutation was aneuploid.

Patient B underwent 1 cycle; 18 oocytes were collected, of which 17 were fertilized. Fourteen zygotes developed into blastocysts and were biopsied. The unfertilized oocyte and 1 embryo, which arrested during blastocyst formation, were tubed whole. Blastomeres of 2 embryos arrested during cleavage divisions were tubed individually. Analysis of the m.8993T>G mutation and PGT-A was performed in all samples. The mutation was not detected in any sample. Ten biopsies were euploid; all other samples were aneuploid. The transfer of 1 male embryo resulted in a pregnancy and delivery of a healthy boy at 40 weeks of gestation. The mutation was not detected in any of the newborn's samples (Supplemental Table 4, available online).

Patient C underwent 2 separate cycles. Three oocytes were collected during her first cycle, and 2 oocytes were fertilized. One zygote developed to the cleavage stage and was biopsied. The unfertilized oocyte and blastomeres from the arrested cleavage-stage embryo were tubed individually. Mutation analysis at position m.10191 was performed in all samples. PGT-A was performed in the cleavage-stage biopsy. The mutation was not detected in any sample. The biopsy was identified as euploid. The embryo was transferred but failed to implant. In the second cycle, 8 oocytes were collected, and 7 oocytes were fertilized. Three zygotes developed to the cleavage stage and were biopsied. Unfertilized oocytes and embryos arrested during cleavage divisions were tubed whole. Again, mutation analysis was performed in all samples and PGT-A in biopsies. The mutation was not detected in any sample. PGT-A revealed that 1 cleavage-stage biopsy was euploid, and the corresponding embryo was transferred. Although implantation occurred, evident by a positive pregnancy test, the pregnancy was lost shortly after. Because the mutation was not detected in any embryo specimen, the patient's transmission risk was estimated as low. Patient C went on to conceive naturally and underwent PND (chorionic villus sampling). No mutant variant at position m.10191 was detected, and a healthy boy was delivered at full term.

Patient D underwent 1 cycle. Twenty oocytes were subjected to intracytoplasmic sperm injection. Nineteen oocytes were fertilized, and 10 zygotes developed to the blastocyst stage. Of these, 9 were biopsied. One blastocyst, developing from an abnormally fertilized zygote, and the unfertilized oocyte were tubed whole. Nine embryos, which arrested during cleavage divisions, were split, and halves were tubed separately. Analysis of the m.3243A>G mutation was performed in all but 1 sample, which failed to amplify. PGT-A was performed in trophectoderm biopsies. The mutation was detected in the unfertilized oocyte, in both parts of 4 arrested cleavage-stage embryos, and in 3 trophectoderm biopsies, of which 2 also were aneuploid. Of the remaining 6 trophectoderm biopsies, 2 were euploid, and the corresponding embryos were transferred consecutively. Although both embryos implanted and fetal heartbeats were seen, the patient miscarried in both transfer cycles at 8 weeks of gestation.

Of note, this particular case presented some interesting and potentially important findings. First, almost identical mutation levels were detected in both halves of arrested cleavage-stage embryos (mean difference ± SD, 0.9% ± 1.3%; range, 0.0%–3.2%). Moreover, when wild-type homoplasmy was present, it was always detected in both embryo parts. Second, approximately two-thirds (12/20) of embryos were likely mutation-free. Third, although not significant ($P = .1570$), detected mutation levels declined steadily during preimplantation development progression (Fig. 4). The highest mutation level was detected in the unfertilized oocyte (37.5%). At the cleavage stage, a mean mutation level of 13.0% ± 18.2% was detected. The lowest mean mutation level was detected in blastocyst-stage specimens (4.4% ± 9.4%). Fewer than half (4/9) of the cleavage-stage embryos had detectable mtDNA mutation levels, ranging from 5.7% to 38.0% (mean ± SD, 29.3% ± 15.7%) among carrier embryos. Fewer than one-third (3/10) of blastocysts were identified as

carriers with a mean mutation level of 14.6% ± 13.3%, ranging from 3.1% to 29.2%.

DISCUSSION

Validation experiments demonstrated that the proposed NGS method is highly accurate and sensitive for mtDNA mutation detection and quantification and can be reliably applied to minute samples, including embryo biopsies. An excellent correlation was achieved between expected and obtained mutant loads in sample mixing experiments and between mutant loads detected in individuals' gDNA and cell samples. Importantly, results were consistent when WGA was performed before the amplification of mutation loci, verifying that WGA does not distort variant ratios. Equally important, despite variation in intercellular mutant loads, which is frequent in mtDNA mutation carriers (11, 23–26), mean cell mutant loads were always close to levels seen in corresponding bulk gDNA samples.

Mutation levels established by reference laboratories were confirmed in samples from m.8993T>G carriers. We did not have access to muscle DNA samples, which were used to establish m.10191T>C and m.3243A>G reference levels, and mutant loads were consistently higher or lower, respectively, in the cheek swab and blood samples (present study). This was not unexpected because m.3243A>G mutant load varies between tissues and tends to be lowest in leucocytes, followed by buccal cells, and then skeletal muscle cells (27, 28). This probably also occurs in patients carrying the m.10191T>C mutation but is yet to be confirmed (29, 30). Of note, m.8993T>G heteroplasmic load is usually relatively uniform across tissue types (31).

Patients A, B, and C were referred for PGT for Leigh syndrome (OMIM: 256000), a progressive neurodegenerative disorder, often causing childhood death (32). To date, ≥ 60 genes implicated in the disease have been identified (33). Those located in the mtDNA cause maternally inherited Leigh syndrome, which accounts for up to one-quarter of Leigh syndrome cases (34). Mutations in the MT-ATP6 gene (mitochondrially encoded ATP synthase membrane subunit 6), which encodes for a subunit of the respiratory chain complex V, are a frequent cause of maternally inherited Leigh syndrome. Of these, the base change T>G at position m.8993 is the most frequent (35).

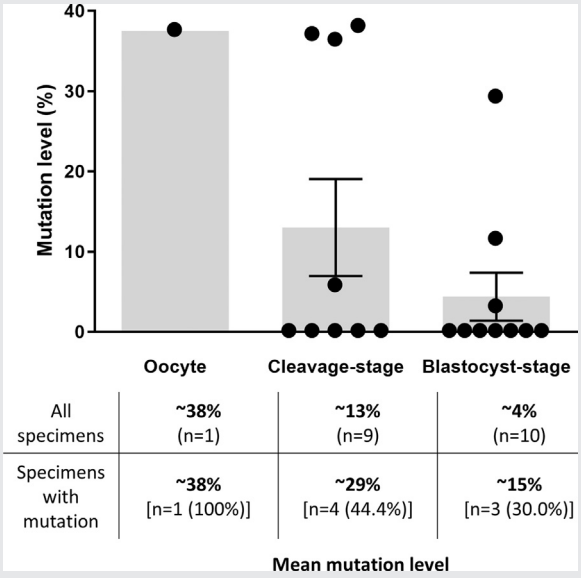
The severity of m.8993T>G is strongly dependent on mutant mtDNA load (33, 36). For instance, the mutant mtDNA load is approximately 56% in Patient A, who is asymptomatic, whereas her severely affected son was homoplasmic for the mutation. Patient A's single competent blastocyst was likely mutation-free but affected by aneuploidy, including trisomy 16, a lethal abnormality common in miscarriages (37). PGT further revealed that 1 of Patient A's arrested embryos was mutation-free, whereas the other and the unfertilized oocyte carried extremely high mutation levels (approximately 90%). Extreme segregation of m.8993T>G mtDNA toward homoplasmy for either wild-type or mutant is common in oocytes and embryos of m.8993T>G carriers (10, 38–40). Although mothers tend to be asymptomatic, severely affected children are often the only family member with maternally inherited Leigh syndrome (36, 41, 42).

Frequently, mothers of affected children are not only asymptomatic but also seemingly mutation-free (42–44). In these cases, mtDNA mutations may arise de novo during germ cell or early embryo development, and the recurrence risk in subsequent pregnancies is low (42). For instance, Patient B's son carried high m.8993T>G mutation levels, but the mutation was undetectable in Patient B, her mother, and in Patient B's oocyte and embryo specimens. Nevertheless, maternal inheritance cannot be ruled out due to the possibility of gonadal mosaicism and the presence of a subpopulation of affected oocytes (42). In this instance, PGT provided information suggestive of a low recurrence risk and ultimately identified an unaffected euploid male embryo that led to the birth of a healthy son. Importantly, genetic testing of the baby confirmed the PGT results.

Similarly, no mutant mtDNA was detected in Patient C or her oocytes or embryos, although high levels of the m.10191T>C mutation were detected in her daughter's muscle biopsy. The m.10191T>C mutation is a rare cause of mitochondrial disease. The limited data available suggest that severity is driven by the extent of heteroplasmic mutant load (29, 45). The m.10191T>C mutation is located in the MT-ND3 gene (mitochondrially encoded NADH:ubiquinone oxidoreductase core subunit 3), which encodes for a subunit of the respiratory chain complex I. Cases of de novo and maternal origin have been reported (45). The absence of detectable mutant mtDNA in any of Patient C's oocyte and embryo specimens supported her initial counseling that the recurrence risk was low (42). Consequently, after 2 failed treatment cycles, the patient opted to conceive naturally. PND confirmed the absence of the mutation, and the patient gave birth to a healthy child.

Unlike the first 3 patients, who were referred for PGT because of affected children, Patient D was referred due to the diagnosis of mitochondrial myopathy caused by m.3243A>G. This mutation, which is located in the MT-TL1 gene (mitochondrially encoded tRNA-Leu [UUA/G] 1), is the most common pathogenic mtDNA mutation (46, 47). It causes a variety of disorders, including maternally inherited diabetes and deafness (OMIM: 520000), mitochondrial myopathy, encephalopathy, lactic acidosis, and stroke-like episodes (MELAS Syndrome; OMIM: 540000), chronic progressive external ophthalmoplegia (OMIM: 590050), myoclonic epilepsy associated with ragged-red fibers (MERRF Syndrome; OMIM: 545000) and other less severe phenotypes (48). Although Patient D generated many embryos, only 2 of 3 euploid blastocysts were identified as likely mutation-free and were transferred. Following counseling, the patient opted not to have the third euploid blastocyst transferred, although relatively low mutation levels (approximately 3%) were detected in the biopsy. Unlike many other pathogenic mtDNA mutations, m.3243A>G heteroplasmy levels are poorly predictive of disease expression, onset, type, and severity (49). Mutation levels often increase postnatally and accumulate over time in somatic tissues (50), and it is, therefore, preferable to transfer embryos that are likely mutation-free. Unfortunately, the patient miscarried twice. Although reasons were

FIGURE 4



Mean m.3243A>G mutation levels detected in embryo specimens at different stages of preimplantation development. Gray bars illustrate the mean mutation levels at different stages of preimplantation development. Each black dot illustrates the mutation level of a single oocyte or embryo. Dots of cleavage-stage embryos refer to mean levels calculated from both embryo halves (also applicable to the cleavage-stage bar). Error bars depict the standard error of the mean. Percentages in bold refer to mean mutation levels. Numerical values and percentages in brackets refer to oocyte or embryo numbers and frequencies.

Spath. mtDNA preimplantation genetic testing. Fertil Steril 2021.

unclear, it should be noted that pregnancy complications, including spontaneous abortion, are not uncommon in women with mitochondrial disease due to m.3243A>G (51, 52).

Analysis of cleavage-stage embryos demonstrated that m.3243A>G mutation levels were uniformly distributed. This finding is important in the context of PGT, as it indicates that mutation levels in biopsy specimens are likely representative of the entire embryo. This agrees with previous studies demonstrating uniform mutation distribution in embryonic cells, irrespective of mutation type (10–12, 15, 25, 40). Interestingly, the mutation did not appear to be randomly distributed among the patient's embryos. Not only were nearly two-thirds of embryos likely mutation-free, but also levels above approximately 40% were not recorded, suggesting that the mutation may not be transmitted via genetic drift alone but selected against during oogenesis. Additionally, mutation levels and frequencies of mutation-carrying embryos appeared to decrease during preimplantation development progression. While it was not possible to confirm this statistically given the small number of samples available, it raises the possibility that high mutation levels may negatively impact embryo development (i.e., contributing to developmental arrest). It is highly likely that mechanisms exist, in addition to random genetic drift, to prevent pathogenic mtDNA mutation transmission (53–55). Although exact types and timings of these mechanisms are partially documented in humans, they have been demonstrated more clearly during mouse oogenesis and embryogenesis (54, 56). Accumulating evidence suggests that pathogenic mutations are selected against during human germline development, oocyte maturation (57, 58), and embryo development (59). There are also clear differences in transmission patterns between pathogenic mutations (11, 60). As mentioned before, m.8993T>G often displays skewing toward wild-type or mutant homoplasmy in oocytes and embryos (10, 38–40), which is reflected by major heteroplasmy level shifts in pedigrees (41). By contrast, oocytes and embryos from m.3243A>G carriers display substantially less variation in mutation levels and an absence of very high levels and mutation homoplasmy (11, 15, 40, 61). It should also be noted that m.3243A>G, which impairs the assembly of several complexes of the ATP-generating machinery, is selected against in rapidly dividing postnatal tissues. By contrast, levels of m.8993T>G, which affects a single complex, remain stable postnatally. Thus, it is highly likely that negative selection also occurs against oocytes and possibly embryos with high m.3243A>G levels, although those carrying m.8993T>G may escape selection (60). However, data on m.3243A>G distribution in embryos are very limited, and a relation with embryo viability has not been reported. Whether the present decline of m.3243A>G during preimplantation development progression is coincidental or truly reflects negative selection only can be established by investigating larger cohorts.

In conclusion, we report the first successful application of an NGS-based PGT approach that combines PGT for mtDNA disease and PGT-A. Aneuploidy affected half of the embryos tested and involved chromosomes frequently associated with implantation failure and miscarriage. The identification and avoidance of transfer of aneuploid embryos are predicted to offer benefits compared with previous methods of PGT for mtDNA disease that take no account of chromosomal status. This study provides useful information for counseling women at risk of transmitting mtDNA disease and highlights the role PGT can potentially play in providing empirical data to support counseling. In agreement with previous findings (42), the recurrence risk in women in whom the pathogenic mutant mtDNA is undetectable was lower than in heteroplasmic women. Lastly, the study findings are consistent with the existence of mutation-specific selective mechanisms influencing mtDNA mutation inheritance.

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Aplicación clínica de métodos basados en secuenciación para pruebas genéticas preimplantacionales paralelas para la enfermedad del ADN mitocondrial y aneuploidía.

Objetivo: Validar y aplicar una estrategia que permita realizar pruebas genéticas preimplantacionales (PGT) paralelas para la enfermedad del ADN mitocondrial (ADNmt) y la aneuploidía (PGT-A).

Diseño: Validación de pruebas preclínicas e informes de casos.

Entorno: Centros de fertilidad. Laboratorio de diagnóstico.

Pacientes: Cuatro pacientes con riesgo de transmitir la enfermedad del mtDNA causada por m.8993T> G (Pacientes A y B), m.10191T> G (Paciente C) y m.3243A> G (Paciente D). Los pacientes A, B y C tenían niños afectados. Los pacientes A y D mostraron heteroplasmia somática para mutaciones del mtDNA.

Intervenciones: Biopsia de embriones, pruebas genéticas y transferencia uterina de embriones que se predice que son euploides y libres de mutaciones.

Principales medidas de resultado: Prueba de precisión, resultados del tratamiento y segregación de mutaciones.

Resultados: Se confirmó la precisión de la cuantificación de la mutación del mtDNA. La prueba fue compatible con PGT-A, y se demostró que la mitad de los embriones analizados eran aneuploides (16/33). Se detectaron mutaciones en aproximadamente el 40% de las biopsias de embriones de las pacientes A y D (10/24), pero en ninguna de las pacientes B y C (n = 29). Las pacientes B y C tuvieron hijos sanos después de PGT y concepción natural, respectivamente. La mutación m.8993T> G mostró segregación sesgada, mientras que los niveles de mutación m.3243A> G fueron relativamente bajos y potencialmente impactaron el desarrollo del embrión.

Conclusiones: Teniendo en cuenta las tasas elevadas de aneuploidías, las estrategias que proporcionan una combinación de PGT para enfermedad del ADNmt y aneuploidía pueden ser ventajosas en comparación con los enfoques que solo consideran el ADNmt. Las mujeres heteroplasmáticas tuvieron una mayor incidencia de embriones afectados que aquellas con ADNmt mutante somático indetectable, pero aún así pudieron producir embriones libres de mutaciones. Aunque no son concluyentes, los resultados son consistentes con la existencia de mecanismos de segregación específicos de la mutación que ocurren durante la ovogénesis y posiblemente la embriogénesis.

Palabras clave: Pruebas genéticas pre-implantacionales, enfermedad del ADN mitocondrial, secuenciación de próxima generación, embriones, mitocondrias